Smart Attendence System

Cryptographic Classroom Bluetooth + 3D Funny Animal AI Liveness + Off-Hours 5-User Campus AI Lab

Submitted To: Head of Department (HOD) & Principal | Pilot Scope: Classroom CSE-301 (60 Students)

REACTION + TASK TIME	UNIQUE 3D ANIMAL TASKS	OFF-HOURS AI LAB	TOTAL PILOT GRANT
2s React + 5s Perform	6.48 Million (Never Repeats)	23 Hrs/Day (Max 5 Users)	Approx. Rs. 4,725

01. CLASSROOM BLUETOOTH HARDWARE

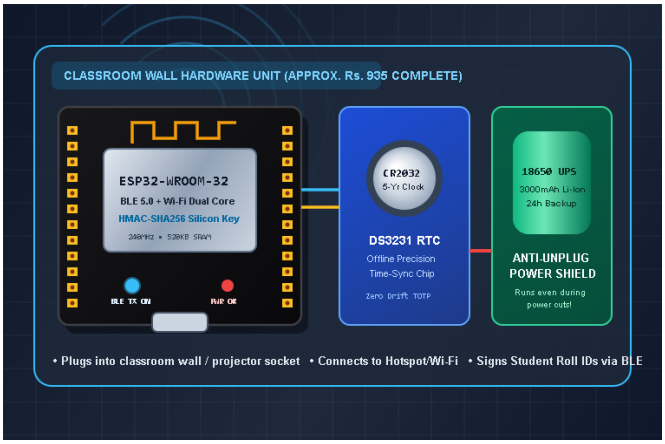
Tamper-Proof ESP32 + RTC + UPS Unit

• Scheduled Window (e.g., 10 PM – 11 PM): Each classroom has one compact ESP32-WROOM-32 box mounted near the podium that activates Bluetooth only during class.

• Zero Clock Drift (DS3231 RTC): Includes a precision real-time clock coin battery module so rotating cryptographic keys stay synced even without internet.

• Anti-Unplug 18650 Battery UPS: Continues working for 24+ hours during power cuts or if someone switches off the wall socket.

• Play Store Distance Check: Our Android app checks Bluetooth signal strength (RSSI) so students must be physically inside the room.



COMPLETE CLASSROOM HARDWARE KIT:

• ESP32 Dual-Radio Board: Approx. Rs. 455

• DS3231 RTC + 18650 UPS + Box: Approx. Rs. 480

Total Per Classroom: Approx. Rs. 935 (vs Rs. 22,000 Biometric)



1:1 IDENTITY LOCK (NO PROXY LOGINS):

Every student is permanently locked 1:1 with (Google+ID, Phone Number+ID, or Apple+ID) + Reference Face Scan + Phone Hardware ID. A friend in class CANNOT log into another student's account.

02. CRYPTOGRAPHIC VPS CONNECTION

How Our VPS Confirms the Classroom Key

• Direct Hotspot/Wi-Fi + 5G Piggyback: The ESP32 connects directly to our VPS over classroom Wi-Fi/Hotspot to sync schedules, and automatically bridges through student 5G phones if Wi-Fi drops.

• Per-Student Hardware Signing (HMAC-SHA256): When Student 22CS104 connects over Bluetooth, the ESP32 signs (Classroom_ID + Roll_22CS104 + Server_Nonce) using its locked silicon secret key.

• Zero WhatsApp Token Sharing: Because the signature is mathematically locked to Roll 22CS104 and expires in seconds, sharing a code with a friend in the hostel is 100% useless.

03. 3D FUNNY ANIMAL LIVENESS ENGINE

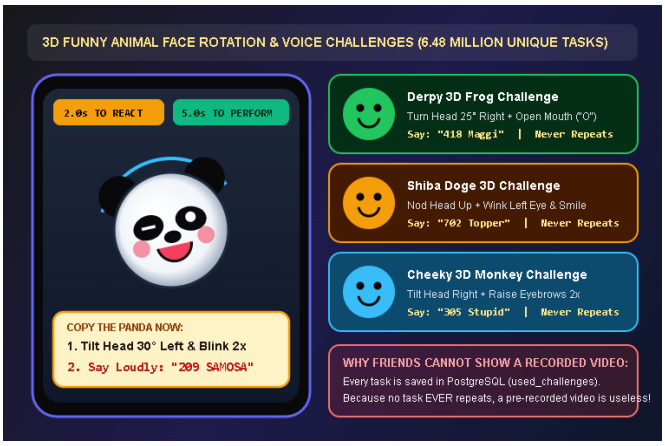
2s to React + 5s to Perform (Never Repeats!)

• Cute & Funny 3D Animal Avatars: Instead of boring geometric shapes, students see an expressive 3D Panda, Derpy Frog, Shiba Doge, or Cheeky Monkey on their screen.

• Mirror the Animal + Speak Random Phrase: The 3D animal rotates its head and performs a random gesture (e.g., "Tilt head left, blink eyes twice, smile, and say 209 Samosa" or "Say Stupid").

• Strict 2s Reaction + 5s Performance Timer: Students have 2 seconds to react and 5 seconds to record the task. Native Android FLAG_SECURE blocks screen recording.

• Saved in Database to NEVER Repeat: Every task combination is hashed and saved in PostgreSQL. If a student records a video of doing a task so a friend can show that video tomorrow, it fails because no task is EVER repeated again in 4 years!



WHY 6.48 MILLION TASKS NEVER REPEAT:

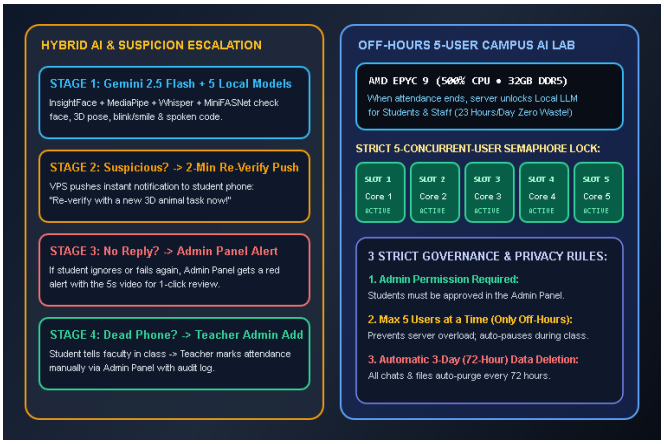
• 6 Expressive 3D Animal Faces (Three.js 60FPS)

• 12 Head Rotations + Eye Blink / Smile Actions

• 900 Spoken Numbers (100-999) + 100 Words

• Stored in PostgreSQL used_challenges table

= 100% Immune to Replay & Video Spoofing!



5 FAST LOCAL AI MODELS ON EPYC 9 VPS:

1. InsightFace (Buffalo_L): 99.4% Face ID Match
2. MediaPipe 478-Mesh: Head Tilt, Blink & Smile
3. Faster-Whisper Audio: Checks Spoken Number
4. MiniFASNet v2: Detects Phone Screen Replay
5. Local Vision Audit LLM + Gemini 2.5 Flash

Verifies all 60 students in ~6 seconds total!

04. AI ESCALATION & 5-USER CAMPUS AI LAB

Suspicion Re-Verify + 23-Hr Student/Staff AI Lab

• Suspicious Attempt -> Instant Student Notification: If our 5 local models or Gemini flag an attempt as suspicious, our VPS immediately sends a Push Notification to the student to re-verify within 2 minutes.

• Unanswered -> Admin Panel Notification: If the student does not reply, the Admin Panel gets an alert with the recorded video so faculty can approve or reject in 1 click.

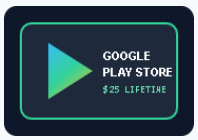
• Manual Teacher Override: If a student's phone is dead or cannot verify, they inform the teacher, who adds their attendance directly via the Admin Panel.

• Off-Hours 5-User Campus AI Lab (Zero Wasted VPS!): When attendance is over, our 32GB DDR5 EPYC 9 server doesn't sit idle—approved students & staff can use the local LLM models for coding & study assistance!

• Strict 5-User Limit & 3-Day Auto-Delete: Requires Admin permission, locks concurrency to max 5 users at a time (only outside attendance hours), and automatically deletes all chat data every 3 days (72 hours).

SMART ATTENDENCE SYSTEM • VISUAL BUDGET BREAKDOWN (1 CLASSROOM PILOT) & SANCTION

Important Budget Note: Because personal Gemini Pro accounts cannot be shared across an entire classroom due to rate limits, a dedicated Gemini 2.5 Flash Pilot Test Credit (Approx. Rs. 455, well below enterprise minimum thresholds) is included for testing alongside our 5 local VPS models. All hardware, VPS, API test credit, and classroom Wi-Fi items are listed as Approximate (Approx.) costs including a 30% buffer for courier delivery charges, cables, and taxes. The Google Play Store Developer registration is kept at its exact Rs. 2,100 ($25 USD) fee without the 30% increase.

VISUAL	COMPONENT, PURPOSE & SPECIFICATION	CYCLE	APPROX. GRANT COST
	**ESP32-WROOM-32 Cryptographic BLE & Wi-Fi Classroom Node** 1× Dual-Core 240MHz MCU with hardware HMAC-SHA256 eFuse & BLE 5.0 GATT server **Includes ~30% delivery, tax & cables buffer (Approx.)**	**One-Time**	**Approx. Rs. 455** (~$5.42 USD)
	**DS3231 RTC Clock + 18650 Mini-UPS Backup + Wall Enclosure & Power Kit** 1× Precision offline clock module, lithium power-cut backup shield, 5V adapter & ABS lockbox **Includes ~30% delivery, tax & cables buffer (Approx.)**	**One-Time**	**Approx. Rs. 480** (~$5.71 USD)
	**Google Play Store Official Developer Console Registration** Official Lifetime Android Play Store publishing license (.aab bundle + FCM Push + FLAG_SECURE) **Official Fixed Google Play Store Fee (0% Markup Applied)**	**One-Time**	**Approx. Rs. 2,100** (~$25 USD)
	**Dedicated AMD EPYC 9 (500% CPU / 5 vCores, 32GB DDR5 RAM) AI & DB Server** Hosts 5 Local AI Verification Models (InsightFace, MediaPipe, Whisper, MiniFASNet), PostgreSQL & 5-Slot Campus AI Lab **Includes ~30% delivery, tax & cables buffer (Approx.)**	**Monthly (Pilot)**	**Approx. Rs. 910** (~$10.83 USD)
	**Gemini 2.5 Flash Multimodal API — Department Pilot Test Credit** Dedicated Pay-As-You-Go trial allocation (~15,000+ multimodal 5-second video/audio verifications) **Includes ~30% delivery, tax & cables buffer (Approx.)**	**Monthly (Pilot)**	**Approx. Rs. 455** (~$5.42 USD)
	**Classroom Wi-Fi / Dedicated Hotspot Telemetry & SSL Domain Connectivity** 1× Dedicated data allocation for live ESP32-to-VPS WebSocket sync, time calibration & HTTPS gateway **Includes ~30% delivery, tax & cables buffer (Approx.)**	**Monthly (Pilot)**	**Approx. Rs. 325** (~$3.87 USD)

1. One-Time Setup Subtotal (ESP32 Node + RTC/UPS/Enclosure + Google Play Store Account): Approx. Rs. 3,035

2. Month 1 Pilot Operations (AMD EPYC 9 32GB DDR5 VPS + Gemini Test Credit + Classroom Wi-Fi): Approx. Rs. 1,690 / month

TOTAL INITIAL GRANT REQUEST (SETUP + MONTH 1 PILOT): Approx. Rs. 4,725 (~$56.25)

Why This Project Delivers 10x Institutional Value for Our College:

1. Saves Rs. 17,000+ Per Room: Commercial biometric scanners cost Rs. 22,000+ per room. Our entire system launches for ~Rs. 4,725.
2. Saves 10 Minutes of Every Lecture: All 60 students verify simultaneously in 7 seconds from their benches.
3. Free 24/7 Campus AI Lab: Gives our department its own private 32GB DDR5 AI Server for student coding & research!

OFFICIAL COLLEGE FUNDING SANCTION & APPROVAL ENDORSEMENT

Sanctioned for pilot deployment in Classroom CSE-301 (60 Students) with an initial grant release of Approx. Rs. 4,725.

Project Lead: B.Tech Project Innovation Team

Approved By: Head of Department (HOD) & Principal